周四（Thursday）**2021.09.16**

P27: 12, P29: 29, P30: 46,P31: 53,54, P32:63

12. In a game of poker, five players are each dealt 5 cards from a 52-card deck. How many ways are there to deal the cards?

5'

29. A poker player is dealt three spades and two hearts. He discards the two hearts and draws two more cards. What is the probability that he draws two more spades?

5'

12. Solution: $\binom{52}{5}\cdot\binom{47}{5}\cdot\binom{42}{5}\cdot\binom{37}{5}\cdot\binom{32}{5}$ 写到这即可.

$$= \frac{52\times 51\times 50\times 49\times 48}{5\times 4\times 3\times 2\times 1}\cdot\frac{47\times 46\times 45\times 44\times 43}{5\times 4\times 3\times 2\times 1}\cdot\cdots\cdot\frac{32\times 31\times 30\times 29\times 28}{5\times 4\times 3\times 2\times 1}$$

$$= 2.976866584\times 10^{30}.$$

29. Solution: $\dfrac{\binom{10}{2}}{\binom{47}{2}} = C_{47}^{10}\cdot C_{46}^{9}$

Other reasonable answers are also acceptable!

Supplementary questions:

9' **1.** Choose 2r shoes from n pairs of shoes with different sizes (2r<n). Compute the probabilities of the following events.

3' a) There are no matching pair among the 2r shoes.

3' b) There are exact ONE matching pair among the 2r shoes.

3' c) Three are r matching pairs among the 2r shoes.

solutions:

a) $P = \dfrac{C_n^{2r} \cdot 2^{2r}}{C_{2n}^{2r}}$

b) $P = \dfrac{C_n^1 C_{n-1}^{2r-2} \cdot 2^{2r-2}}{C_{2n}^{2r}}$

c) $P = \dfrac{C_n^r}{C_{2n}^{2r}}$